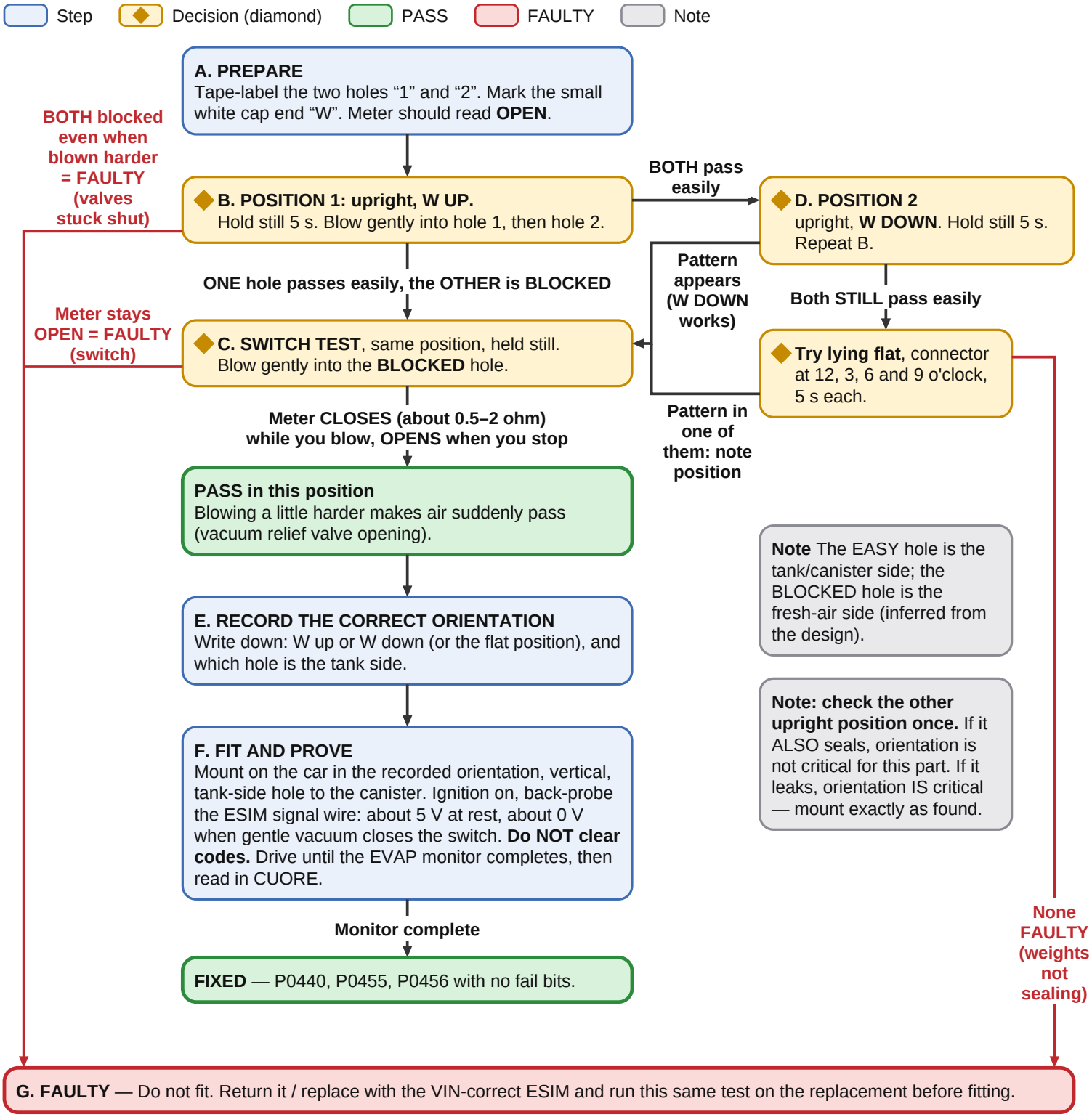
ESIM bench test — new, untested part, out of the car

2018 Alfa Romeo Stelvio 2.0T · Evaporative System Integrity Module · find the correct orientation (small white cap UP or DOWN) and prove the part works

RULES **Blow gently** through a short clean hose — **never suck** (fuel vapour). Gentle = enough to make a candle flicker, not go out. **No battery, no 12 V** — the ESIM has no solenoid. Multimeter on the two switch pins, set to **ohms**. **Hold the ESIM perfectly still**; after every change of position **wait 5 seconds** before blowing.



ESIM bench test — extra checks and sources

WHITE CAP CHECK (optional, if the white cap is visible)

Hold still in the working position and blow gently into the **EASY** hole. The cap should lift and drop back freely the moment you stop. **Stays up or sticks = FAULTY.**

Note: the white cap's identity is not documented — most likely the large weighted pressure valve; rely on the blow and switch results, not on the cap alone. A rattle or click when shaken is NOT a valid test.

WHAT THIS TEST CANNOT PROVE

Breath is not measured, so it cannot confirm the exact switch point (about 1 inch of water vacuum). For that, use a hand vacuum pump with a gauge in inches of water and stay between 1 and 2 inH_2O (above about 2.2 inH_2O the vacuum relief valve opens by design and mimics a leak).

SOURCES (outside literature — none is a reading from this car)

- **Chrysler ESIM design and thresholds** (pressure valve ~0.5 inH_2O, vacuum valve ~2.2 inH_2O, switch ~1 inH_2O, ~0.5–2 ohm closed): vehicleservicepros.com article 53084355 “Chrysler ESIM”.

- **Vertical mounting, no solenoid, 5 V reference:** B&W/BWD tech tip bw10235esim.

- **Wrong orientation causing P0440/P0441/P0456, fixed by re-orienting (Ram):** ALLDATA tech tip “incorrect-eism-install-ram-evaporative-issues”.

Flow directions in the mouth test are inferred from the design.